# Homework 6

Lucy Gagnon

## Question 2

*Please complete the censored* ***Gibbs sampling problem from Module 7, Part II****. You may assume that* $a = 1, b = 1, r = 10$*. You may use any code that was given to you in class or you may choose to write your own code. If you use the code from class, please cite and reference this.*

```
## I used code from class

knitr::opts_chunk$set(cache=TRUE)

# Samples from a truncated gamma with
# truncation (t, infty), shape a, and rate b
# Input: t,a,b
# Output: truncated Gamma(a,b)
sampleTrunGamma <- function(t, a, b){
  p0 <- pgamma(t, shape = a, rate = b)
  # Use the modification of the inverse CD method
  x <- runif(1, min = p0, max = 1)
  y <- qgamma(x, shape = a, rate = b)
  return(y)
}
```

```
# Gibbs sampler
# z is the fully observe data
# c is censored data
# n.iter is number of iterations
# init.theta and init.miss are initial values for sampler
# r,a, and b are fixed parameters
# burnin is number of iterations to use as burnin
sampleGibbs <-
  function(z, c, n.iter, init.theta, init.miss, r, a, b, burnin = 1){
```

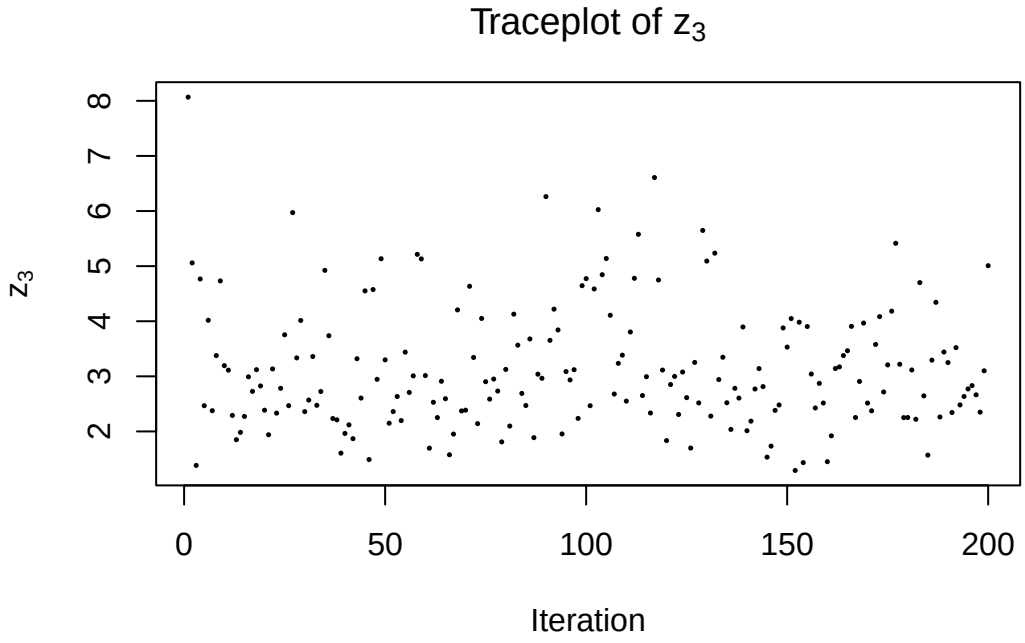
```
  z.sum <- sum(z); m <- length(c); n <- length(z) + m
  miss.vals <- init.miss
  res <- matrix(NA, nrow = n.iter, ncol = 1 + m)
  for (i in 1:n.iter){
    var.sum <- z.sum + sum(miss.vals)
    theta <- rgamma(1, shape = a + n*r, rate = b + var.sum)
    miss.vals <- sapply(c, function(x) {sampleTrunGamma(x, r, theta)})
    res[i,] <- c(theta, miss.vals)
  }
  return(res[burnin:n.iter,])
}
```

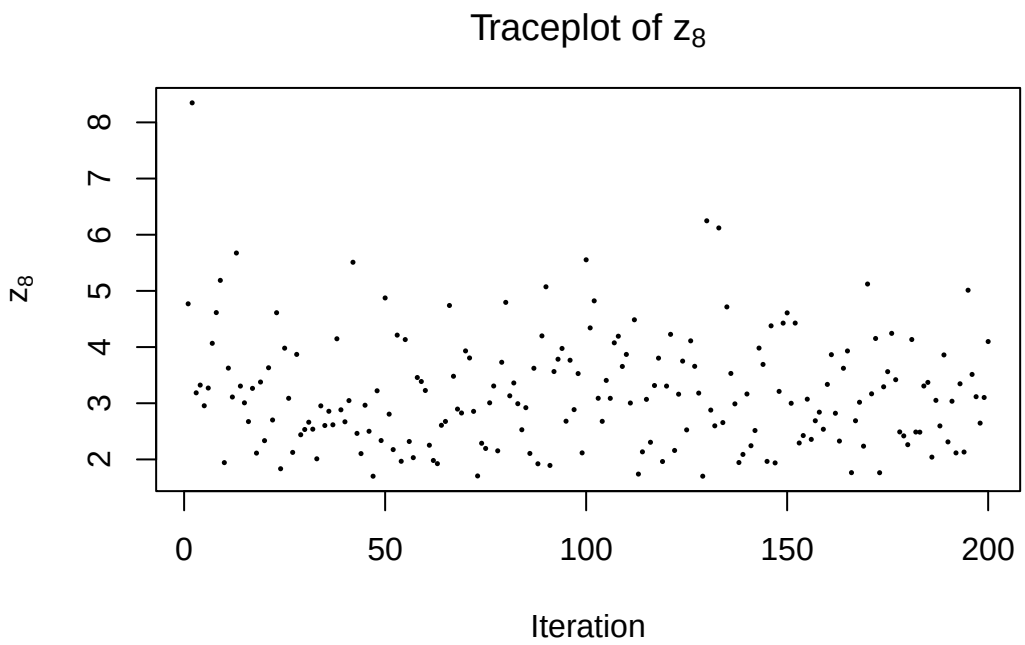
```
set.seed(5983)
# set parameter values and enter data
r <- 10
a <- 1
b <- 1
z <- c(3.4,2.9,1.4,3.2,1.8,4.6,2.8)
c <- c(1.2,1.7,2.0,1.4,0.6)
n.iter <- 200 # changed this to 200 iters ot match Part A
init.theta <- 1
init.missing <-
  rgamma(length(c), shape = r, rate = init.theta)
```

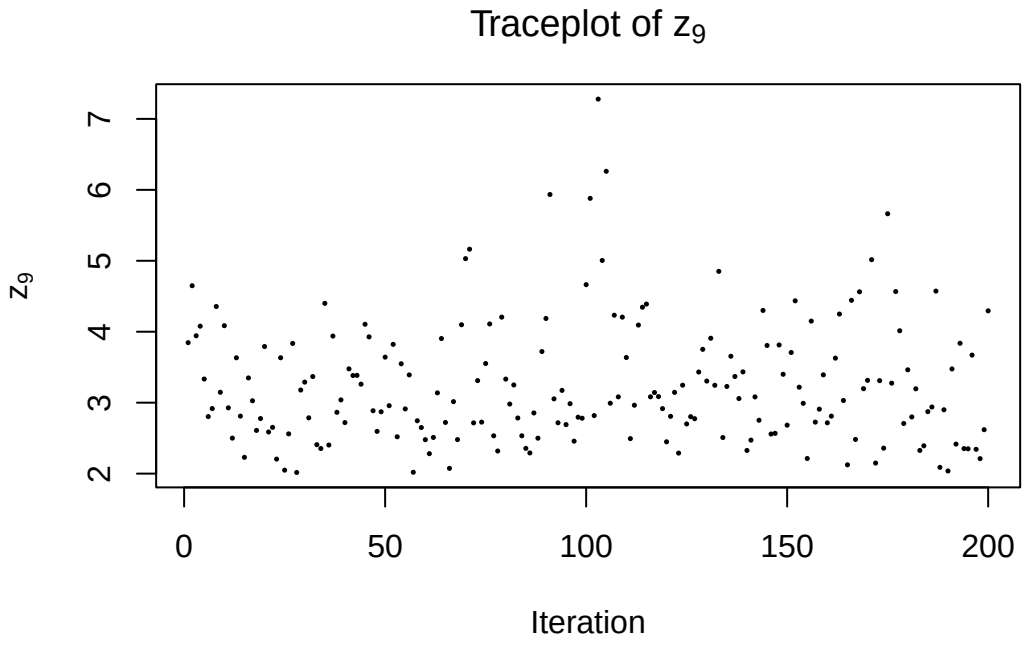
```
# run the Gibbs sampler
res <- sampleGibbs(z, c, n.iter, init.theta, init.missing,
                   r, a, b)
```

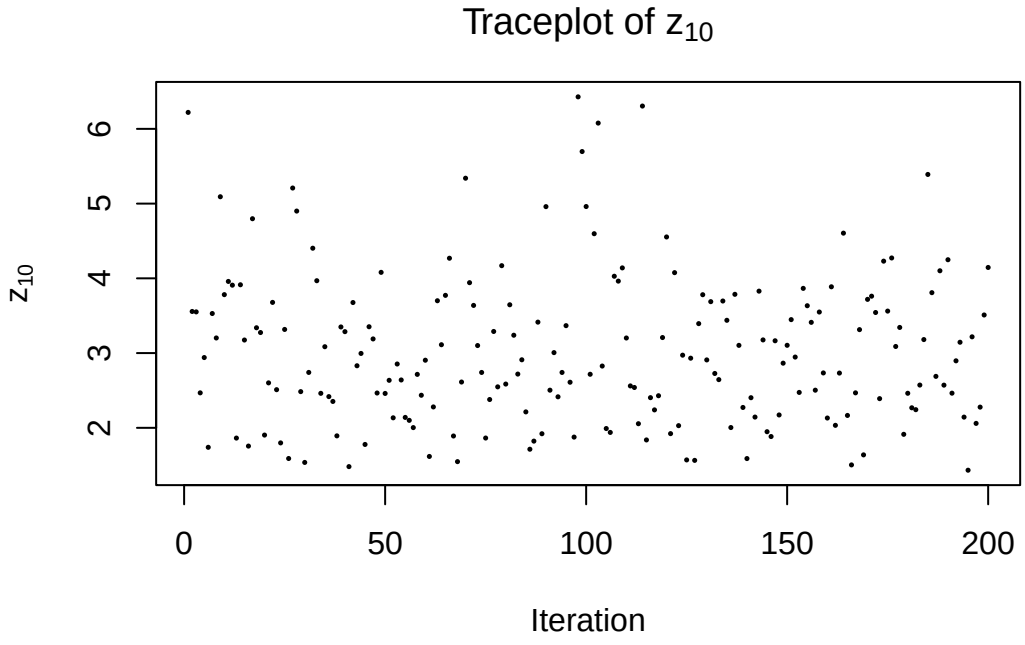
### Part A

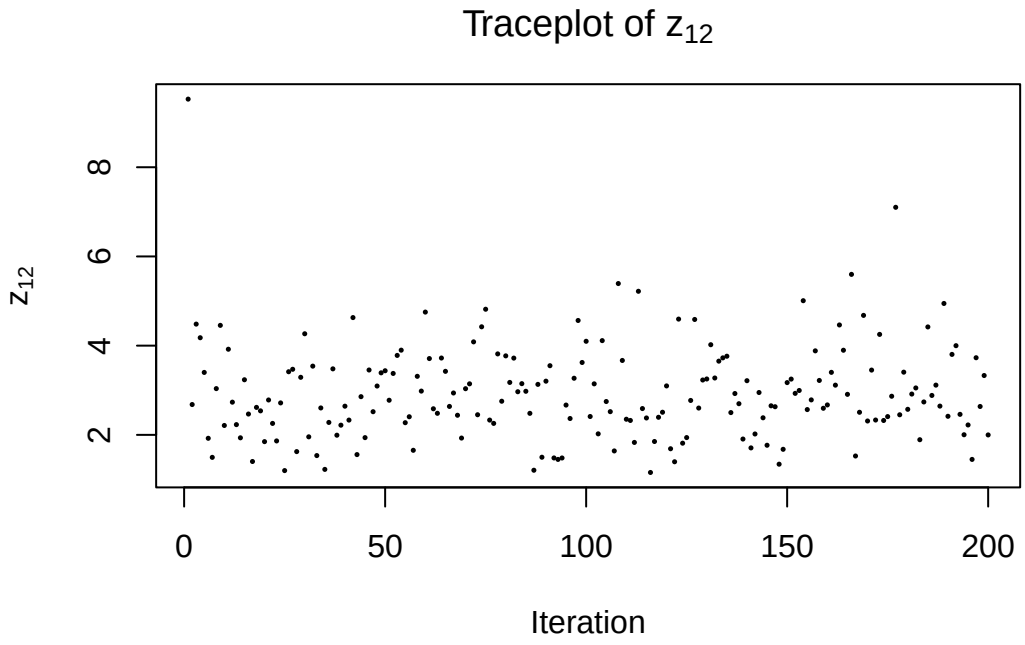
Write code to produce trace plots and running average plots for the **censored values** for 200 iterations. Do these diagnostic plots suggest that you have run the sampler long enough? Explain.

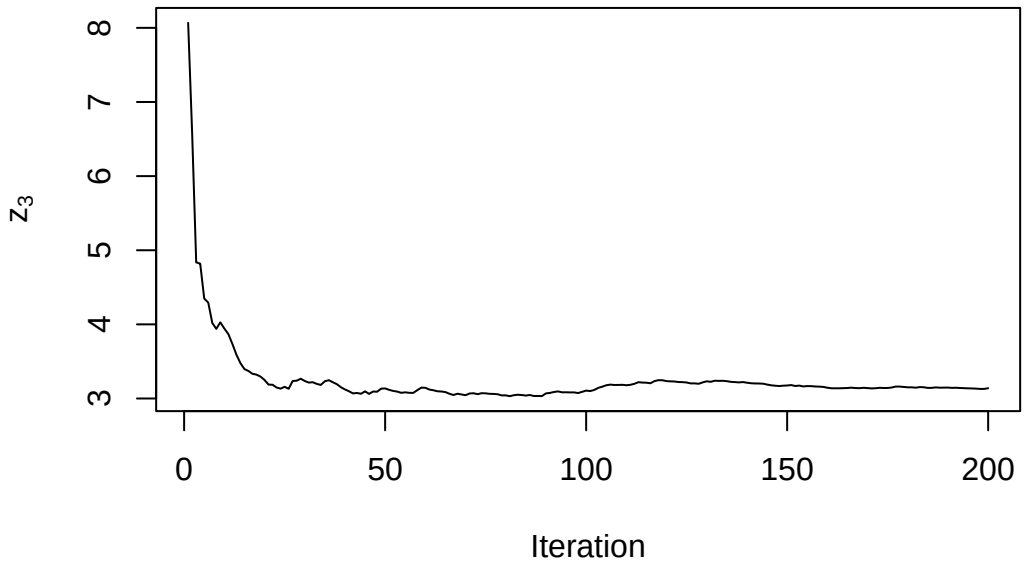

Traceplot of $z_3$
$z_3$
2
3
4
5
6
7
8
0
50
100
150
200
Iteration


Traceplot of $z_8$
$z_8$
2
3
4
5
6
7
8
0
50
100
150
200
Iteration

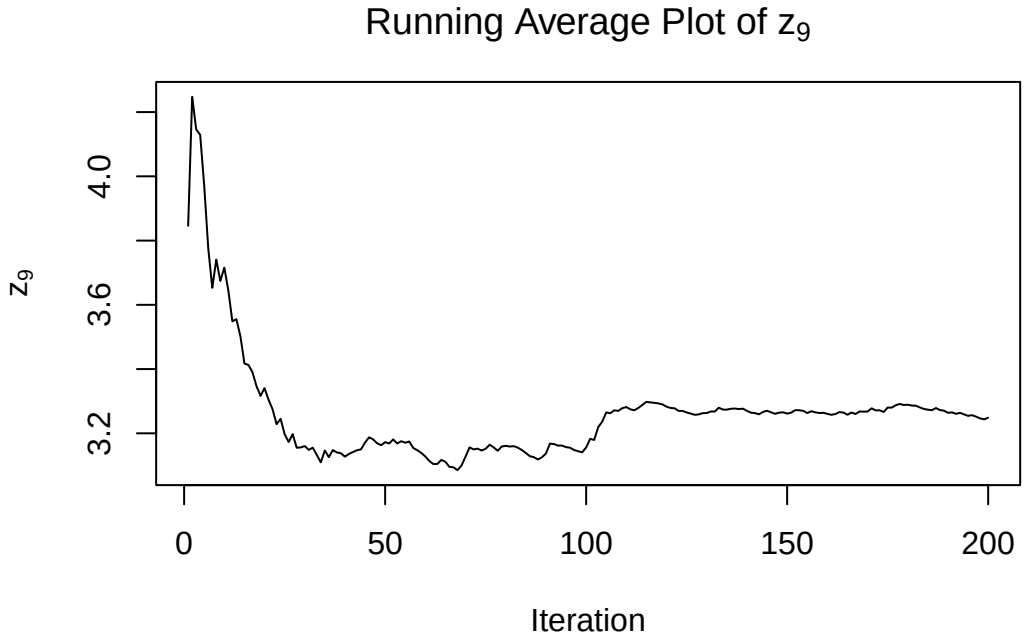

Traceplot of $z_9$
$z_9$
2
3
4
5
6
7
0
50
100
150
200
Iteration

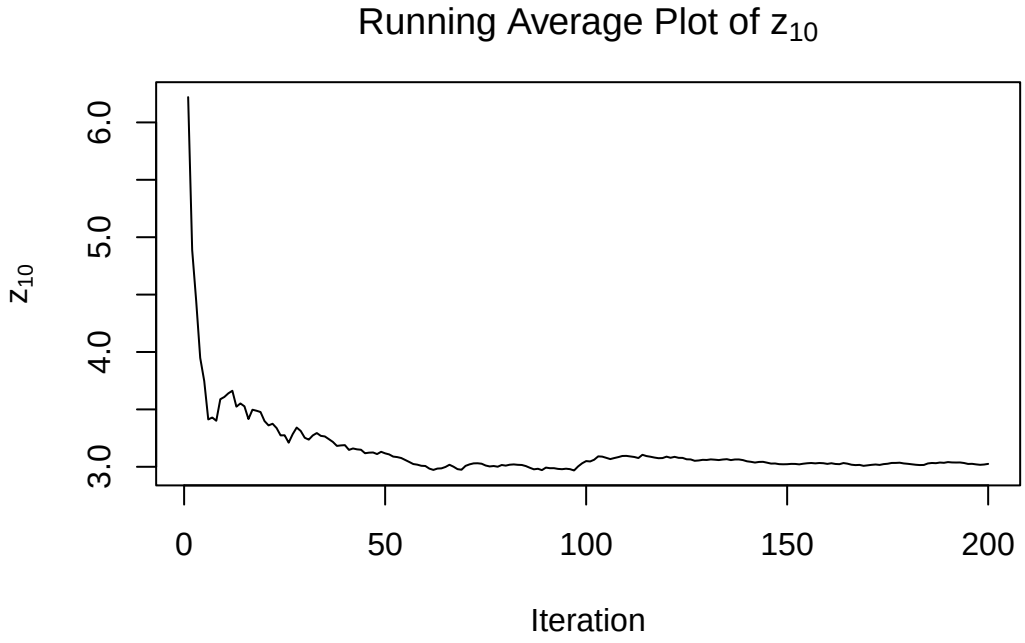

Traceplot of $z_{10}$
$z_{10}$
2
3
4
5
6
0
50
100
150
200
Iteration

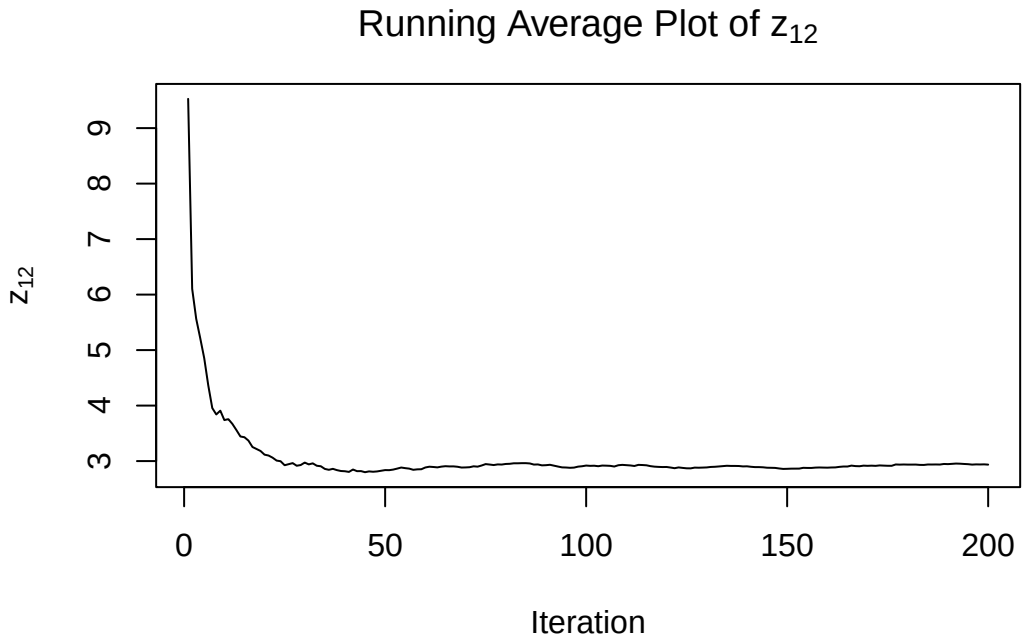

Traceplot of $z_{12}$
$z_{12}$
2
4
6
8
0
50
100
150
200
Iteration


Running Average Plot of $z_3$
$z_3$
3
4
5
6
7
8
0
50
100
150
200
Iteration


Running Average Plot of $z_8$
$z_8$
3.0
4.0
5.0
6.0
0
50
100
150
200
Iteration


Running Average Plot of $z_9$
$z_9$
3.2
3.6
4.0
0
50
100
150
200
Iteration



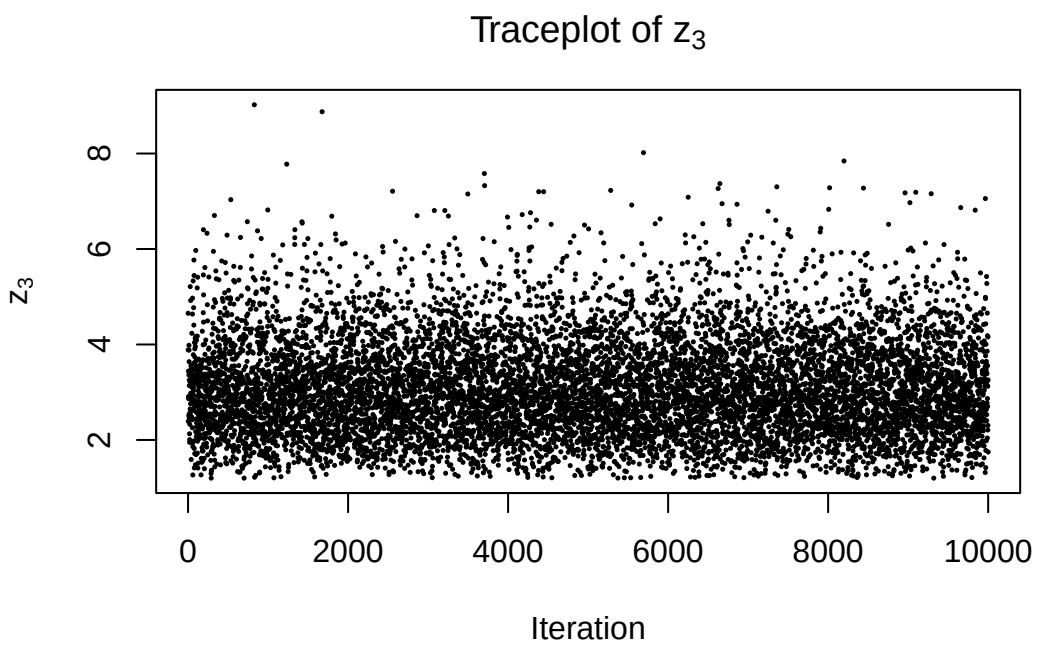



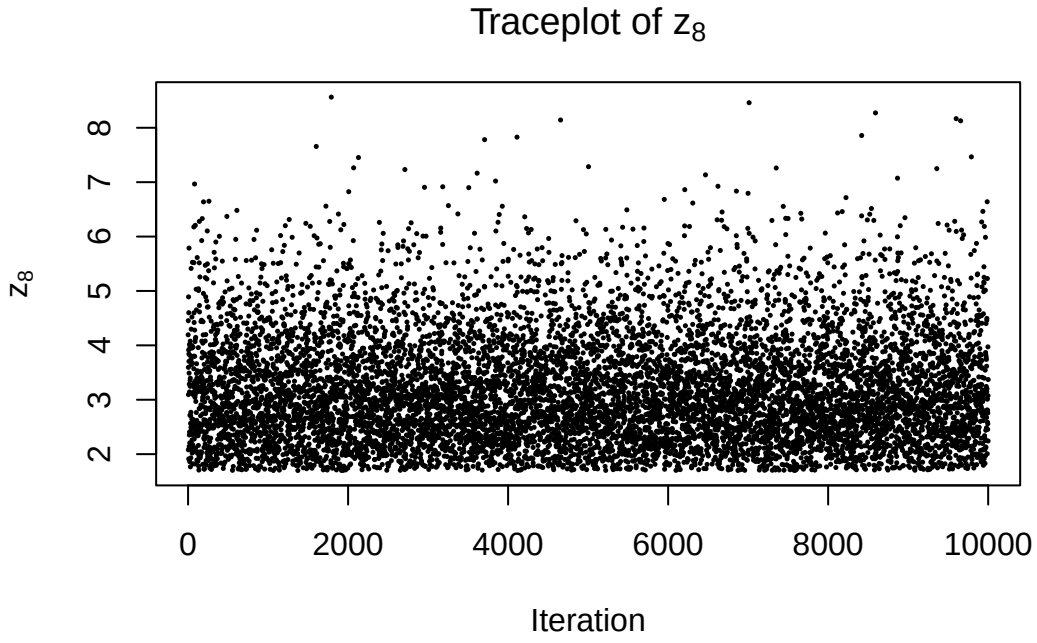

Trace plots and running averages are useful heuristic for visually assessing convergence of a Markov chain. Trace plots may suggest convergence of the sampler when there appears to be good mixing between the sequences of samples and should ideally appear as "noise" or a "bar code" type appearance. Here, the trace plots for the censored values show some "stickiness"

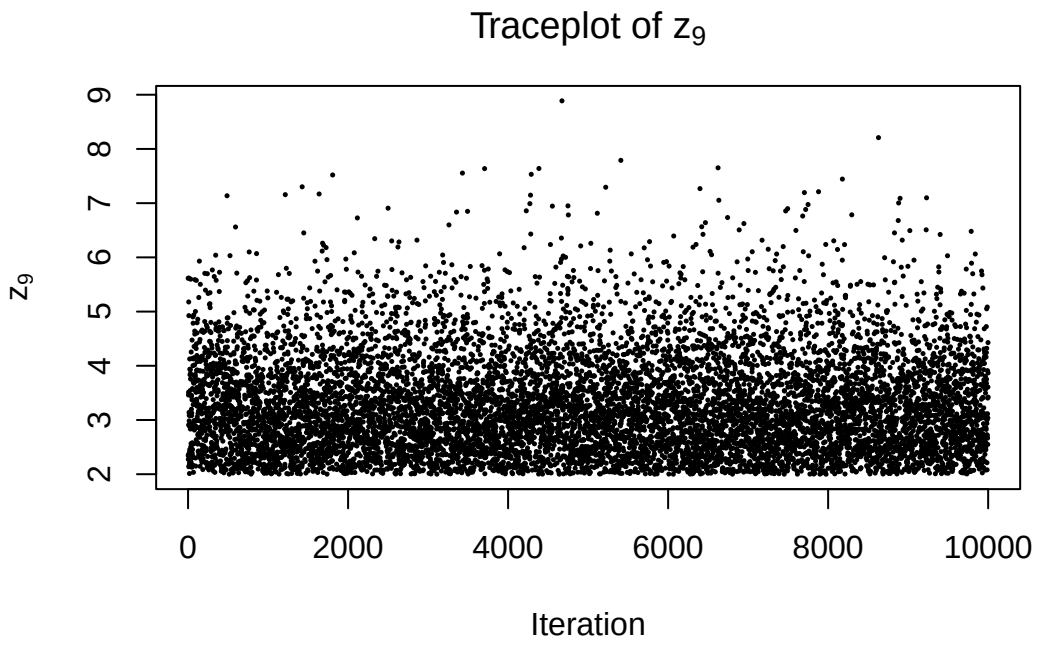
and do not appear as completely "random noise", suggesting we may want to run the sampler for more iterations to achieve results that support convergence. Additionally, the running average plots also support running the sampler for more iterations, as the expectation does not appear to have stabilized across all plots for the censored value samples.

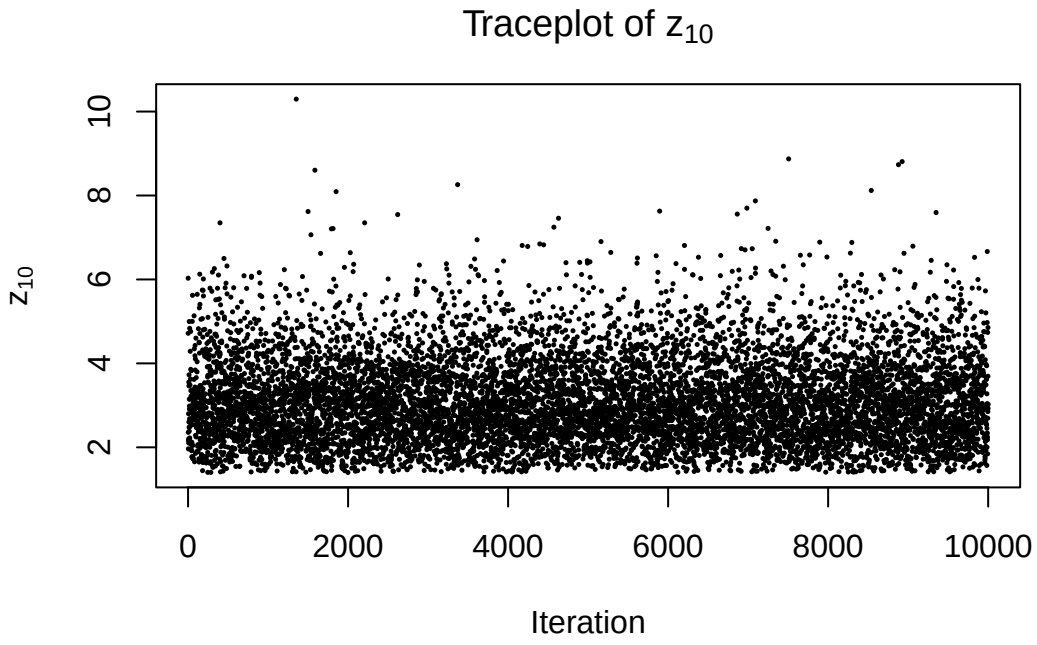
**Part B**

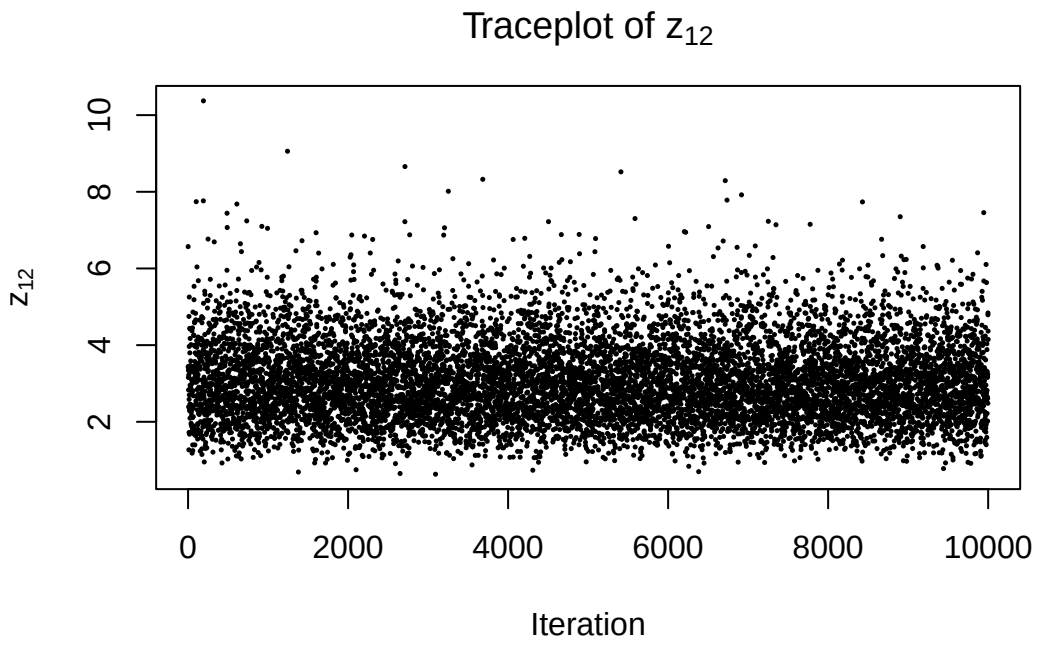
*Now run the chain for 10,000 iterations and update your diagnostic plots (trace plots and running average plots). Report your findings for both trace plots and the running average plots for $\theta$ and the censored values. Do these diagnostic plots suggest that you have run the sampler long enough? Explain.*

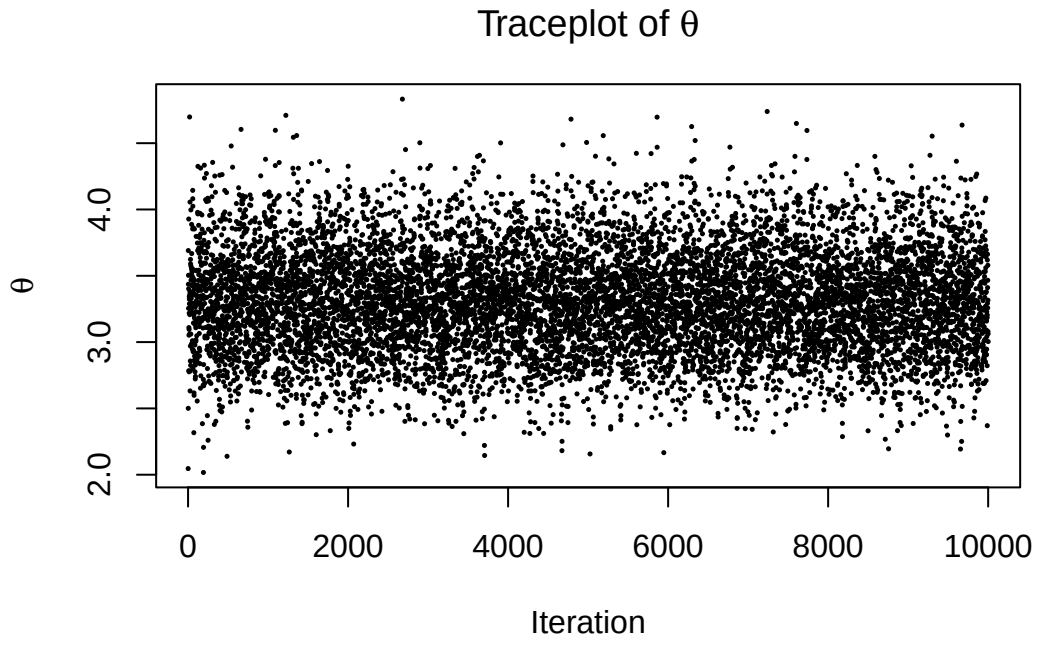


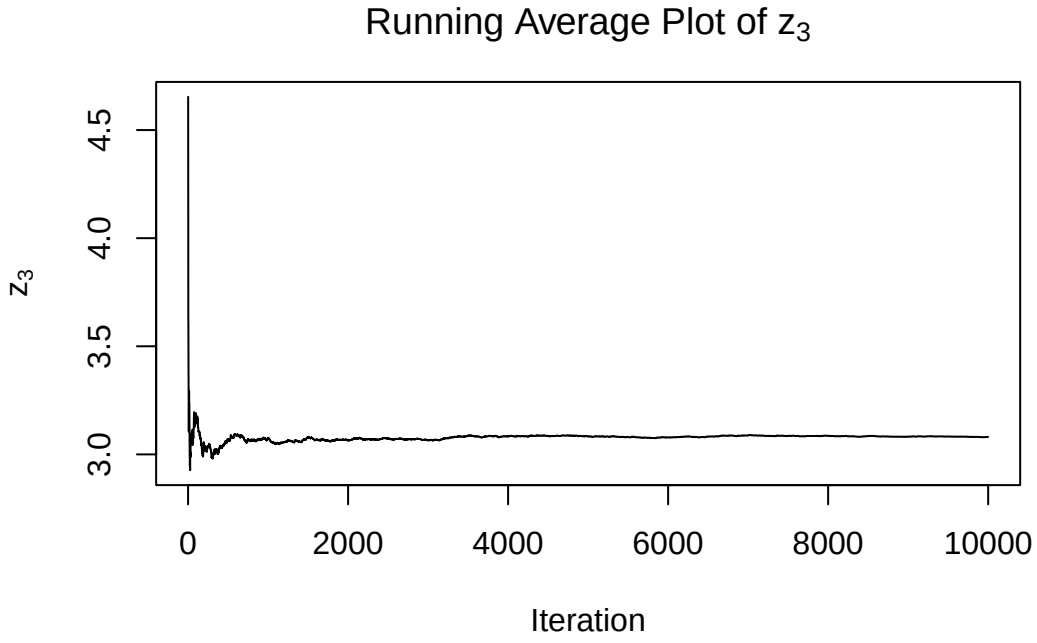

Traceplot of $z_8$
$z_8$
2
3
4
5
6
7
8
0
2000
4000
6000
8000
10000
Iteration

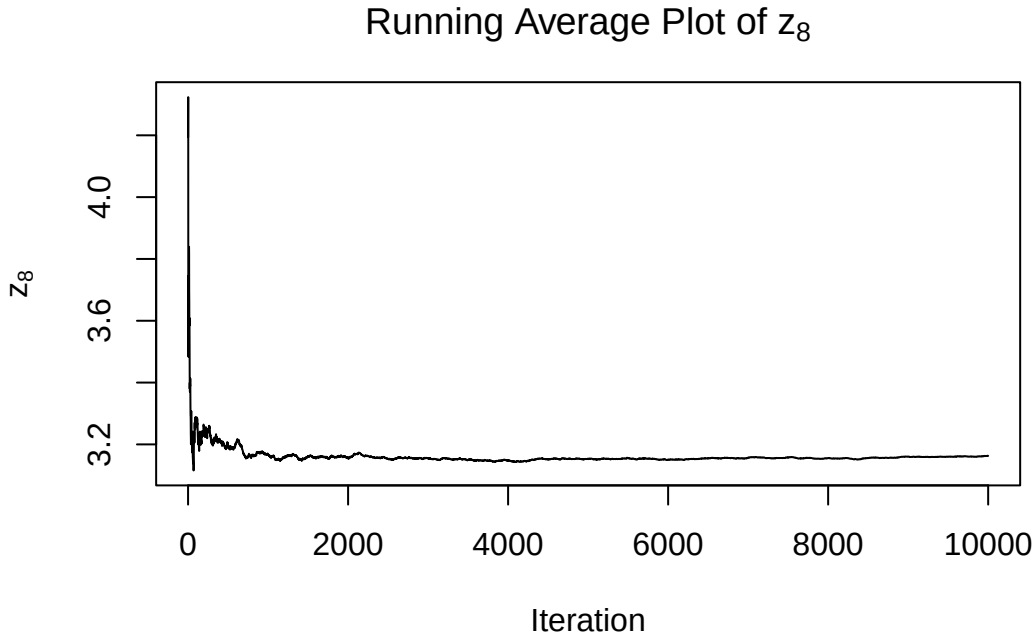

Traceplot of $z_9$
$z_9$
2
3
4
5
6
7
8
9
0
2000
4000
6000
8000
10000
Iteration

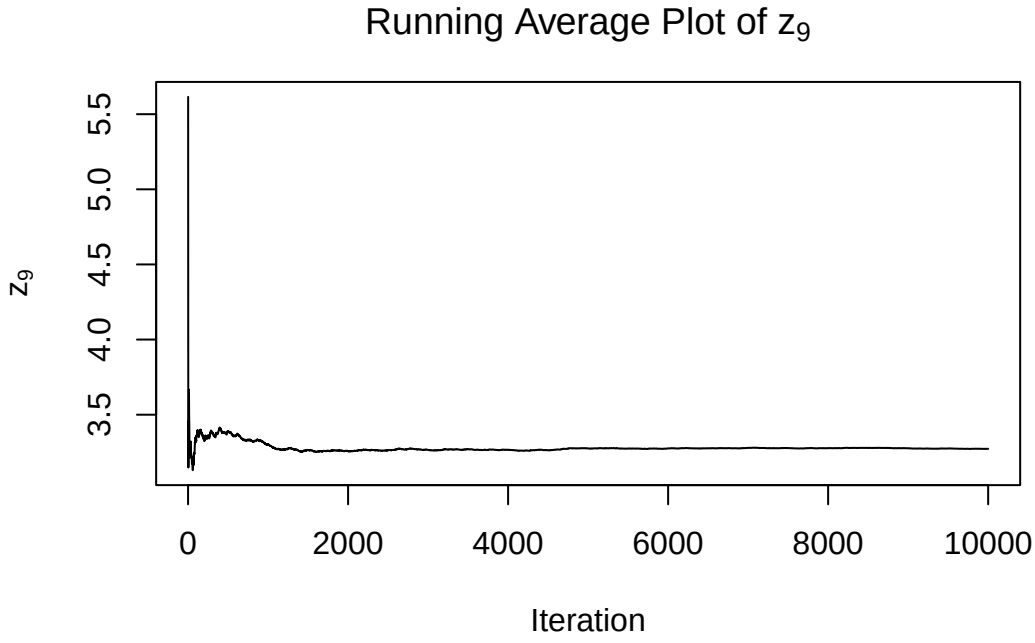

Traceplot of $z_{10}$
$z_{10}$
2
4
6
8
10
0
2000
4000
6000
8000
10000
Iteration


Traceplot of $z_{12}$
$z_{12}$
2
4
6
8
10
0
2000
4000
6000
8000
10000
Iteration


Traceplot of θ
θ
2.0
3.0
4.0
0
2000
4000
6000
8000
10000
Iteration

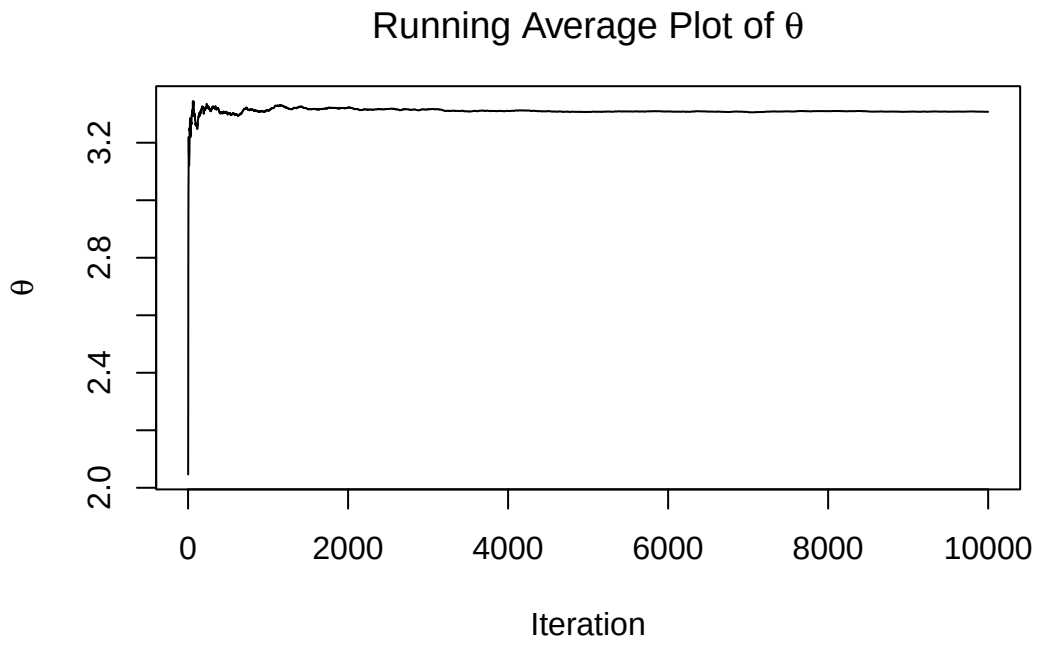

Running Average Plot of $z_3$
$z_3$
3.0
3.5
4.0
4.5
0
2000
4000
6000
8000
10000
Iteration


Running Average Plot of $z_8$
$z_8$
3.2
3.6
4.0
0
2000
4000
6000
8000
10000
Iteration


Running Average Plot of $z_9$
$z_9$
3.5
4.0
4.5
5.0
5.5
0
2000
4000
6000
8000
10000
Iteration


Running Average Plot of $z_{10}$
$z_{10}$
3.0
4.0
5.0
6.0
0
2000
4000
6000
8000
10000
Iteration


Running Average Plot of $z_{12}$
$z_{12}$
3
4
5
6
0
2000
4000
6000
8000
10000
Iteration




When we run the sampler for 10,000 iterations the trace plots for both the censored values and $\theta$ show good mixing/random noise, supporting convergence. Additionally, the running average plots for the censored values and $\theta$ appear to stabilize, supporting convergence.

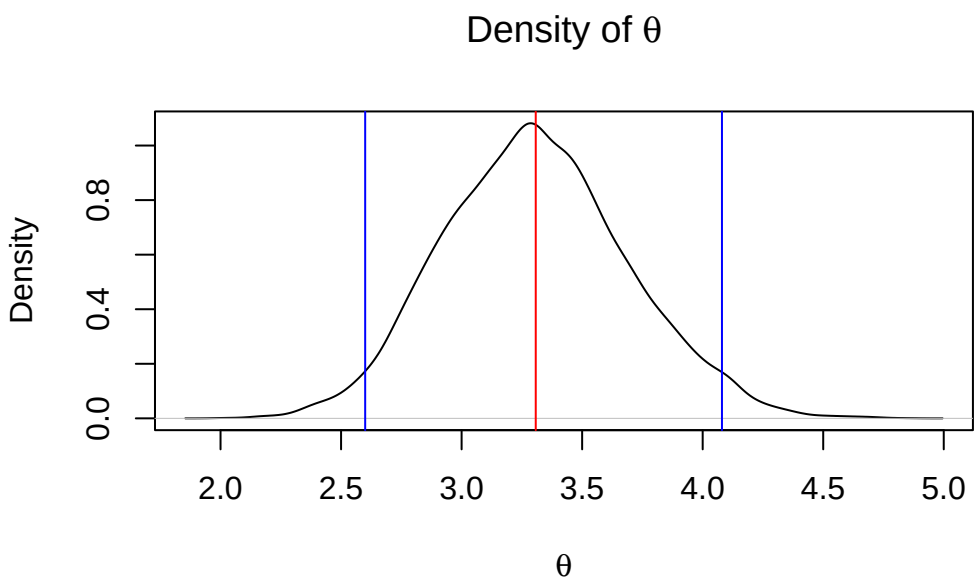
### Part C

*Give plots of the estimated density of* $\theta \mid \cdots$ *and* $z_9 \mid \cdots$*. Be sure to give brief explanations of your results and findings. (Present plots for 10,000 iterations).*

```
# estimated density plot of theta
plot(density(res.new[,1]), xlab = expression(theta),
     main = expression(paste("Density of ", theta)))
abline(v = mean(res.new[,1]), col = "red")
quantile(res.new[,1], c(0.025, 0.975))
```

```
    2.5%    97.5%
2.599349 4.093658
```

```
abline(v = 2.60, col = "blue")
abline(v = 4.08, col = "blue")
```




```
# estimate density plot of censored values
## modified class code
### used ChatGPT to debug error

par(mfrow = c(1, 1), mar = c(4, 4, 3, 1))
missing.index <- c(3, 8, 9, 10, 12)

for (ind in missing.index) {
  x.lab <- bquote(z[.(ind)])
  draws <- res.new[, which(missing.index == ind) + 1]

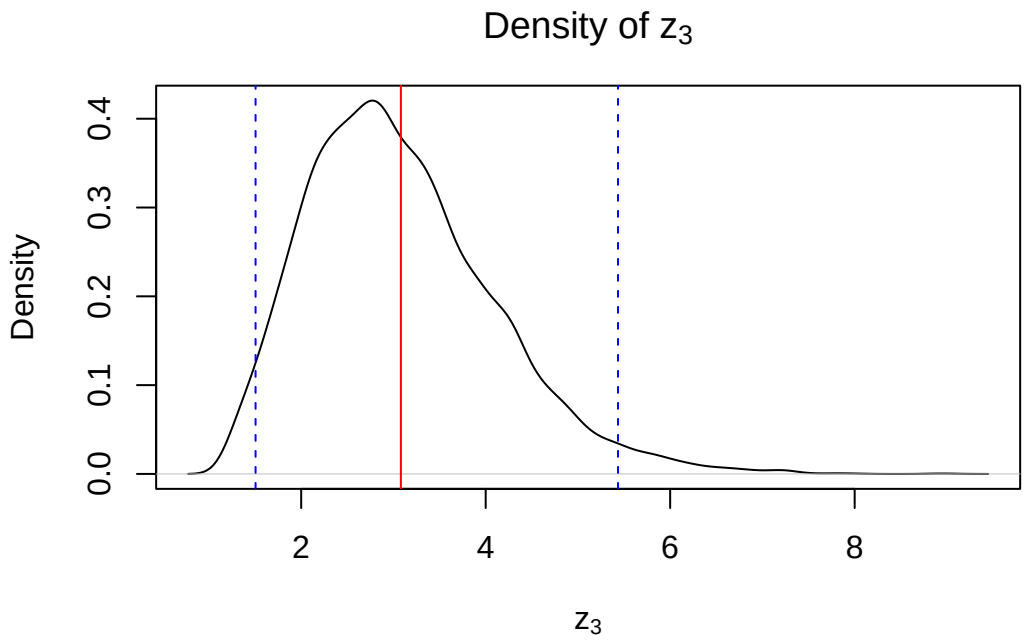
  plot(density(draws),
       xlab = x.lab, ylab = "Density",
       main = bquote(paste("Density of ", .(x.lab))))

  # posterior mean
  abline(v = mean(draws), col = "red")

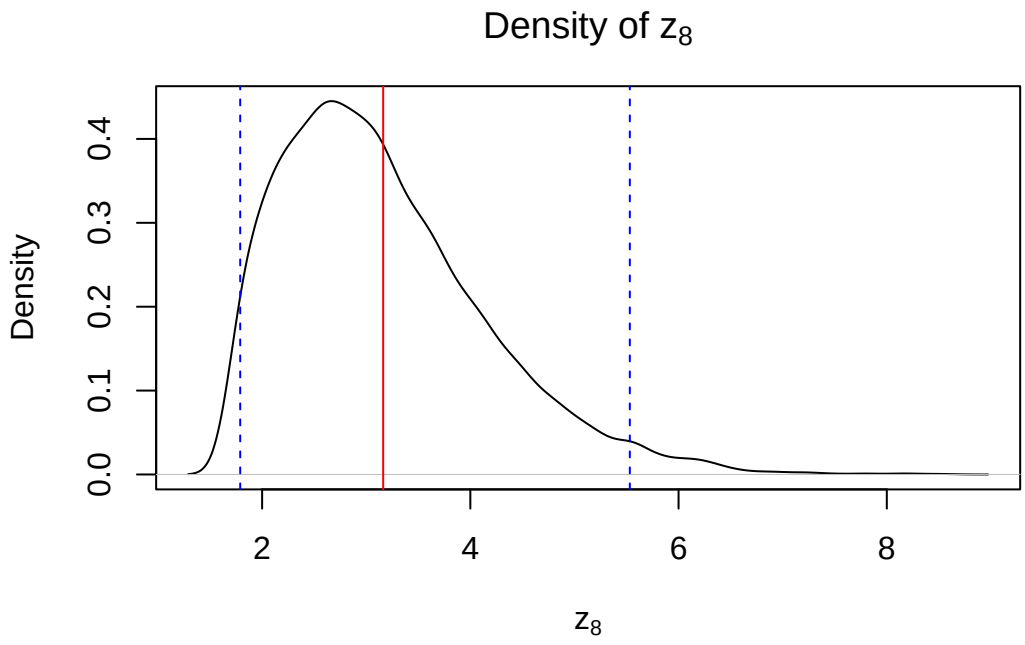
  # equal-tailed 95% credible interval
  ci <- quantile(draws, c(0.025, 0.975))
  abline(v = ci, col = "blue", lty = 2)

  print(ci)
```

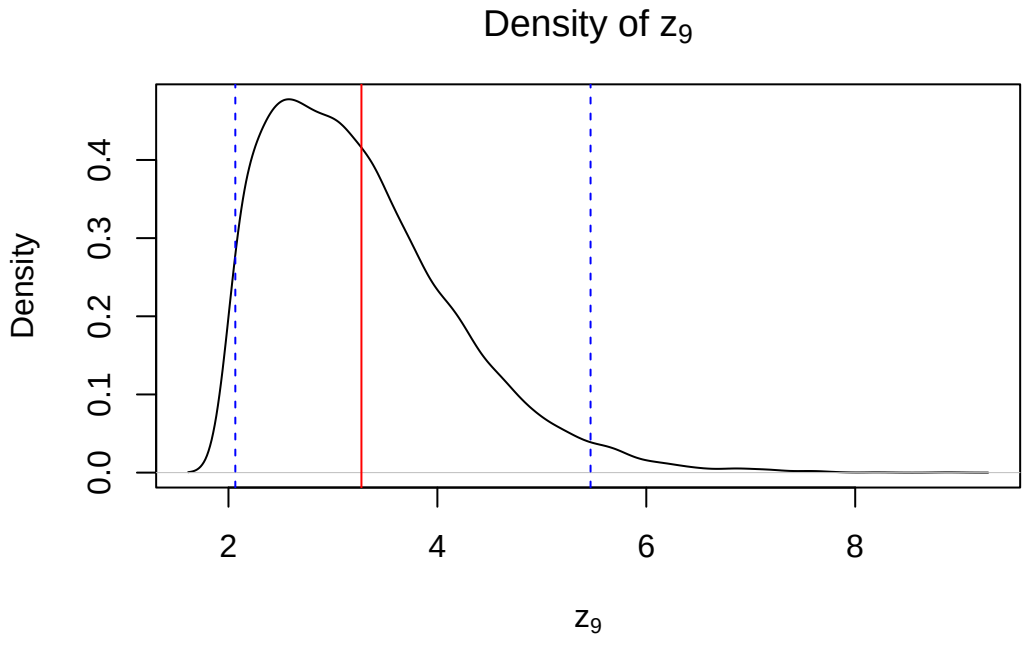
```
}
```



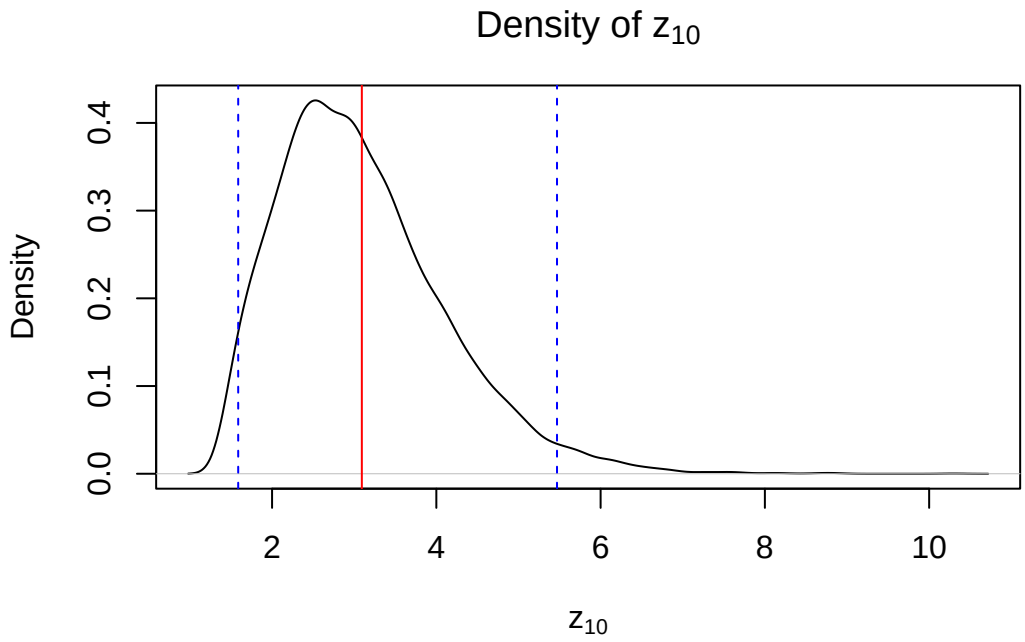

```
    2.5%    97.5% 
1.505371 5.435083 
```



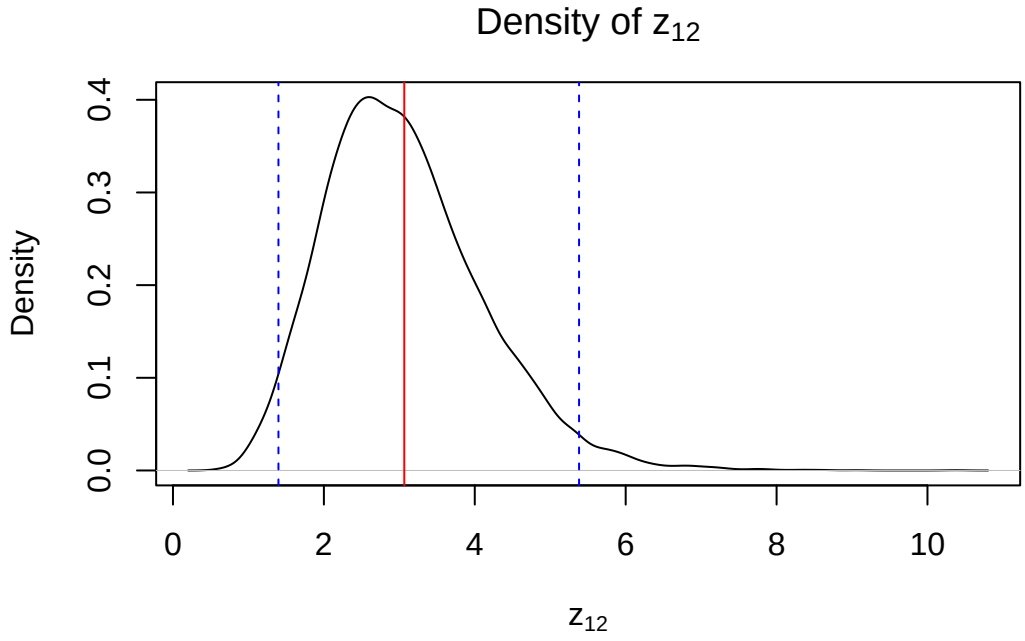

```
    2.5%    97.5%
1.789503 5.531738
```



```
    2.5%    97.5%
2.065696 5.466440
```




```
    2.5%    97.5%
1.587752 5.469382
```




```
    2.5%    97.5%
1.397744 5.382221
```

```
par(mfrow = c(1, 1))
```

Here, we plotted the estimated density of $\theta \mid \cdots$ and $z_9 \mid \cdots$ for 10,000 iterations. The estimated density of theta is the approximated posterior density of the rate of how long individuals are likely to live conditional on all the other parameters, where the red line is the posterior mean of this rate and the blue lines represent a 95 percent credible region. Thus, with shape $r$ fixed, larger values imply shorter expected lifetimes. The estimated density appears relatively smooth and the posterior mean overlaps closely with the mode of the estimated density. The 95% CI ranges from about 2.60 to 4.08, suggesting a positive association between rate of how long individuals are likely to live conditional on all the other parameters and actual lifetime.

Next we plotted the approximated posterior density of the censored random variables conditional on all the other parameters. In contrast, the censored values show a skew right estimated density, so the posterior mean tends to be sligthly greater than their modes. Each distribution represents uncertainty about an individual's total lifetime after intervention. The estimated densities appear relatively smooth, and all 95% CIs have lower bounds greater then 1 and upper bounds less than 6, suggesting that we expect patients with censored data to have a total lifetime after intervention between between about 1 to 6 years.

### Part D

*Finally, let's suppose that* $r = 10, a = 1, b = 100$. *Do the posterior densities in part (c) change for* $\theta \mid \cdots$ *and* $z_9 \mid \cdots$? *Do the associated posterior densities change when* $r = 10, a = 100, b = 1$? *Please provide plots and an explanation to back up your answer. (Use 10,000 iterations for the Gibbs sampler).*

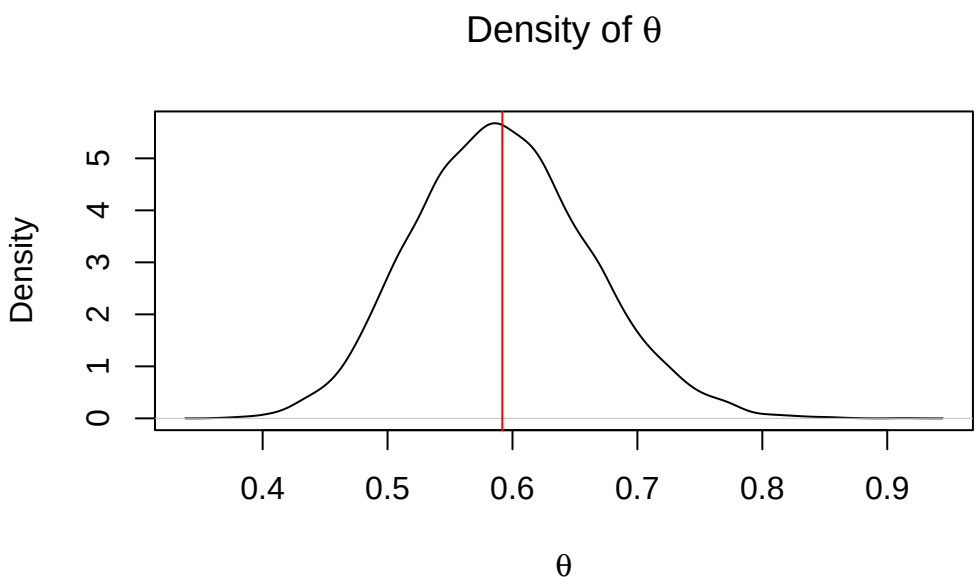
```
# set new parameter values and enter data
b.new <- 100

init.theta <- 1
init.missing <-
  rgamma(length(c), shape = r, rate = init.theta)
# run the Gibbs sampler
res.d <- sampleGibbs(z, c, n.iter.new, init.theta, init.missing,
                     r, a, b.new)

# estimated density plot of theta
plot(density(res.d[,1]), xlab = expression(theta),
     main = expression(paste("Density of ", theta)))
abline(v = mean(res.d[,1]), col = "red")
quantile(res.d[,1], c(0.025, 0.975))
```

```
     2.5%     97.5%
0.4635228 0.7362235
```

```
abline(v = 6.30, col = "blue")
abline(v = 8.36, col = "blue")
```




```
# estimate density plot of censored values
## modified class code
### used ChatGPT to debug error

par(mfrow = c(1, 1), mar = c(4, 4, 3, 1))
missing.index <- c(3, 8, 9, 10, 12)

for (ind in missing.index) {
  x.lab <- bquote(z[.(ind)])
  draws <- res.d[, which(missing.index == ind) + 1]

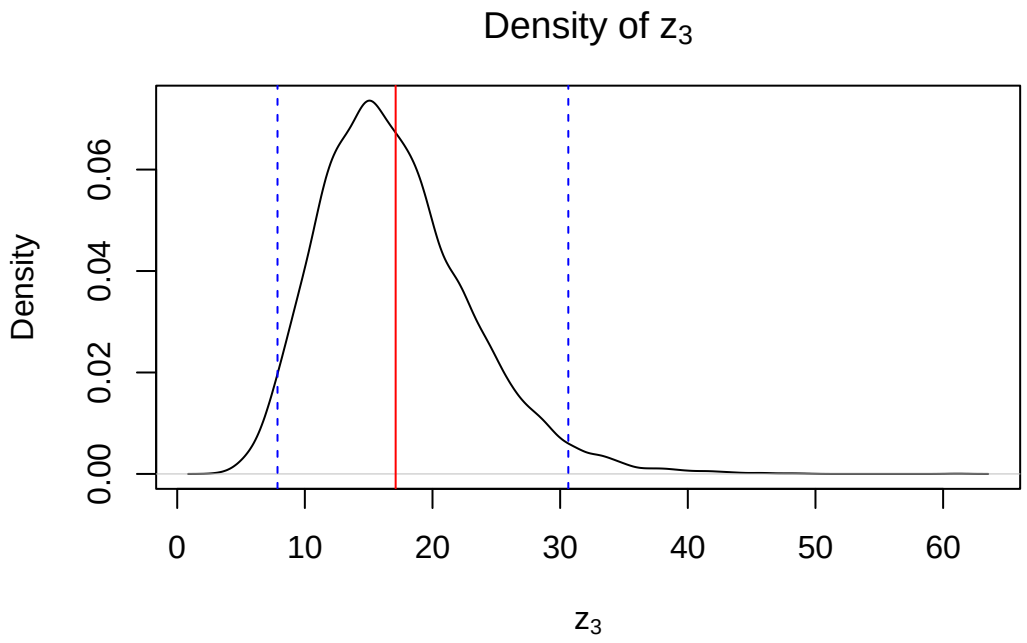
  plot(density(draws),
       xlab = x.lab, ylab = "Density",
       main = bquote(paste("Density of ", .(x.lab))))

  # posterior mean
  abline(v = mean(draws), col = "red")

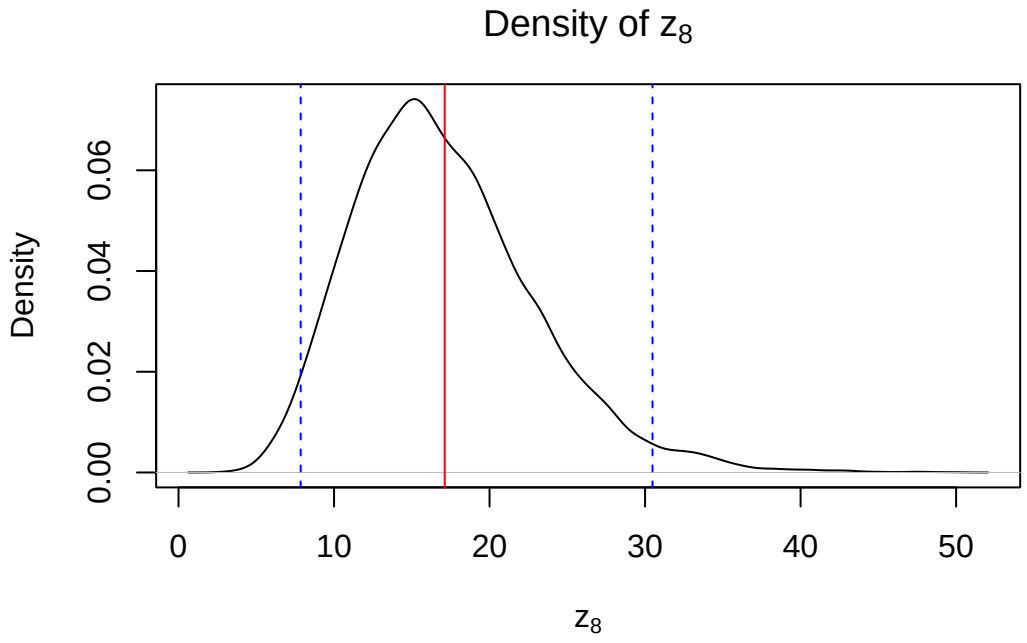
  # equal-tailed 95% credible interval
  ci <- quantile(draws, c(0.025, 0.975))
  abline(v = ci, col = "blue", lty = 2)

  print(ci)
```

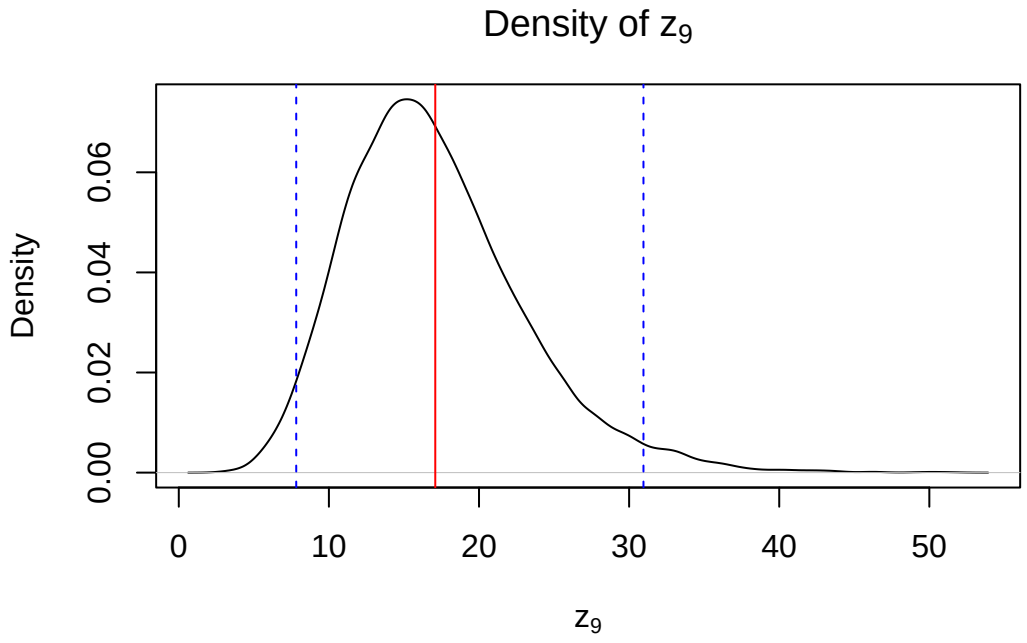
```
}
```



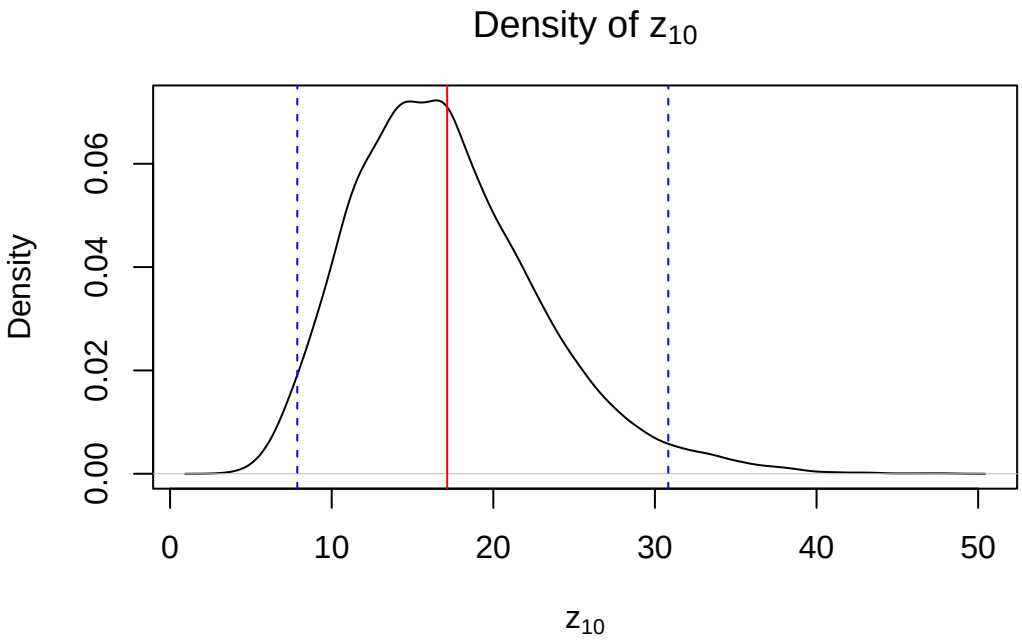

```
    2.5%     97.5% 
7.857953 30.638010 
```



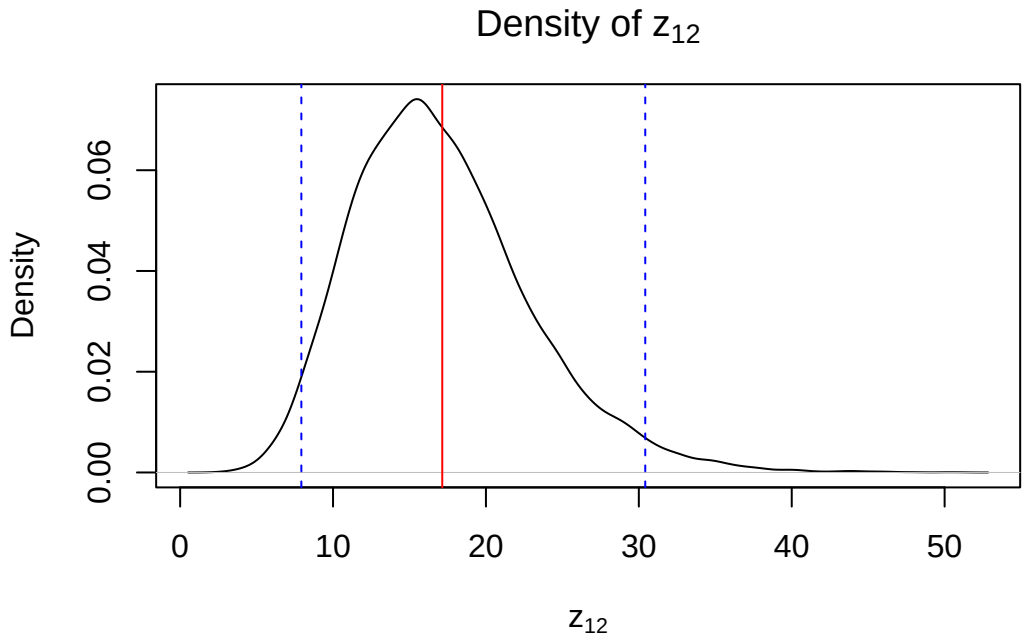

```
    2.5%    97.5%
7.85601 30.47773
```



```
    2.5%     97.5%
7.822964 30.955659
```




```
    2.5%     97.5%
7.871695 30.828142
```




```
     2.5%     97.5%
 7.927423 30.420793
```

```
par(mfrow = c(1, 1), mar = c(4, 4, 3, 1))

# set new parameter values and enter data
a.new <- 100

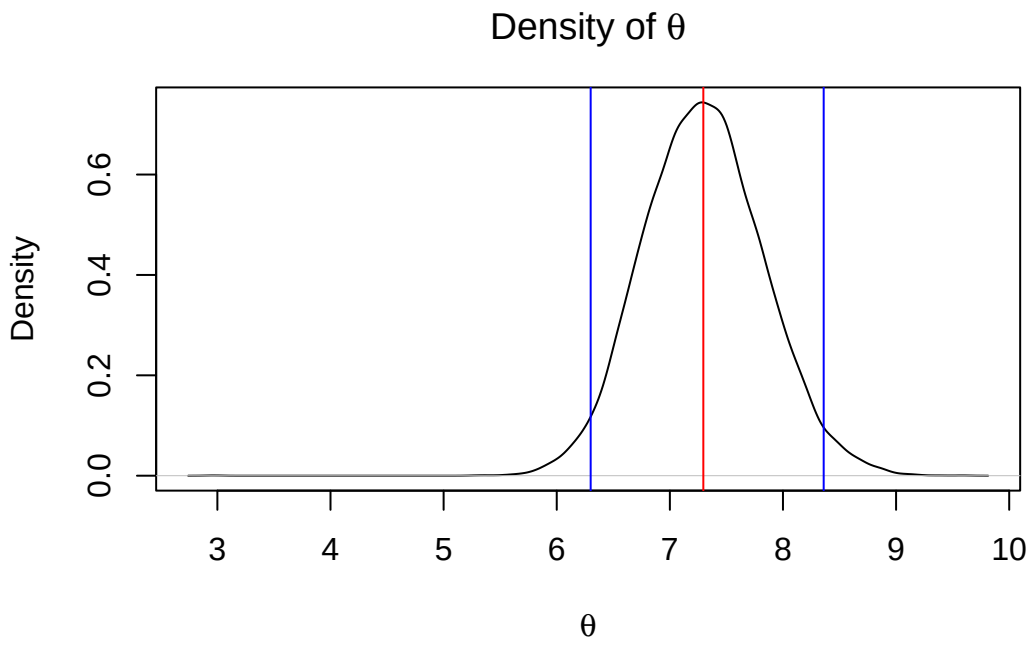
init.theta <- 1
init.missing <-
  rgamma(length(c), shape = r, rate = init.theta)
# run the Gibbs sampler
res.d <- sampleGibbs(z, c, n.iter.new, init.theta, init.missing,
                     r, a.new, b)

# estimated density plot of theta
plot(density(res.d[,1]), xlab = expression(theta),
     main = expression(paste("Density of ", theta)))
abline(v = mean(res.d[,1]), col = "red")
quantile(res.d[,1], c(0.025, 0.975))
```

```
    2.5%    97.5%
```

```
6.296094 8.361260
```

```
abline(v = 6.30, col = "blue")
abline(v = 8.36, col = "blue")
```




```
# estimate density plot of censored values
## modified class code
### used ChatGPT to debug error

par(mfrow = c(1, 1), mar = c(4, 4, 3, 1))
missing.index <- c(3, 8, 9, 10, 12)

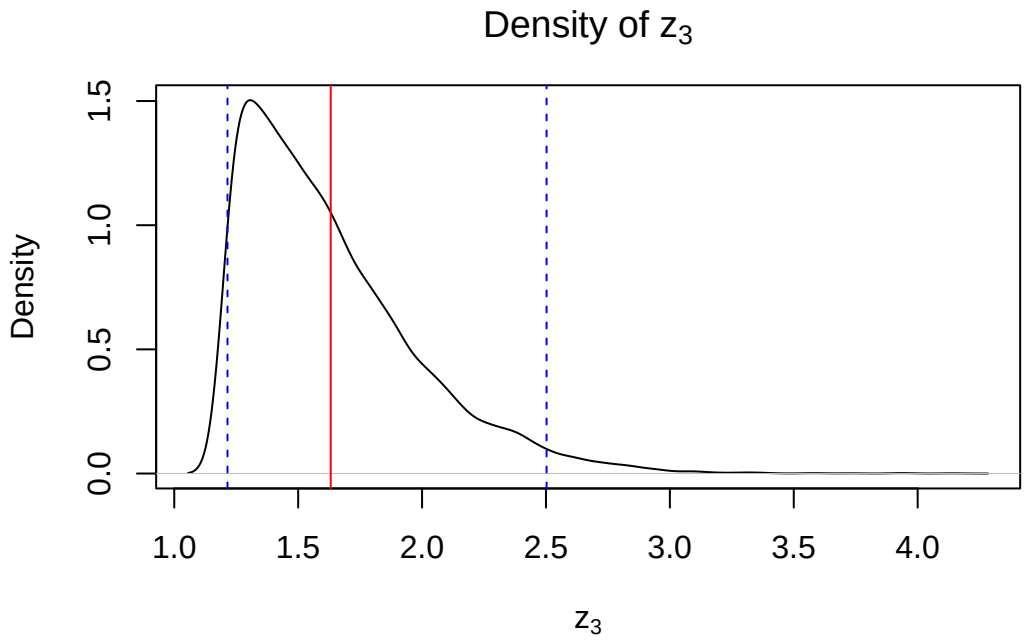
for (ind in missing.index) {
  x.lab <- bquote(z[.(ind)])
  draws <- res.d[, which(missing.index == ind) + 1]

  plot(density(draws),
       xlab = x.lab, ylab = "Density",
       main = bquote(paste("Density of ", .(x.lab))))

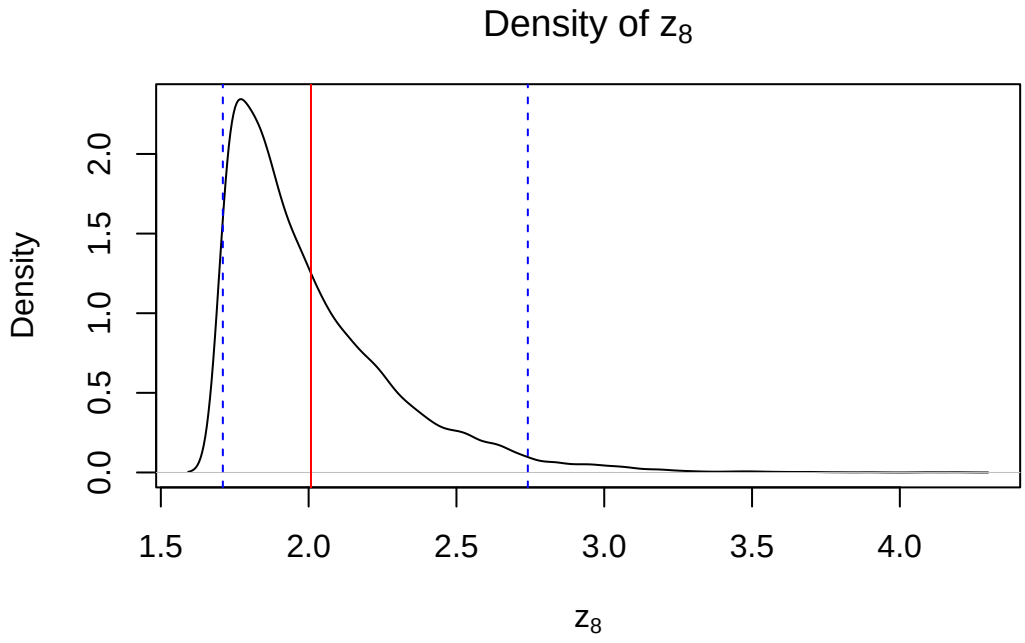
  # posterior mean
  abline(v = mean(draws), col = "red")
```

```r
  # equal-tailed 95% credible interval
  ci <- quantile(draws, c(0.025, 0.975))
  abline(v = ci, col = "blue", lty = 2)

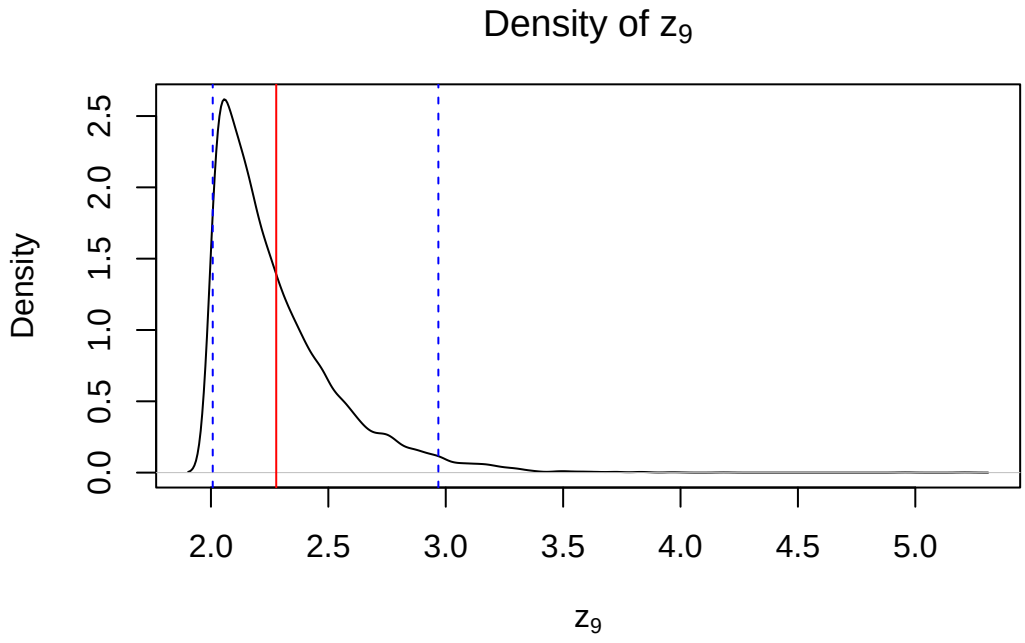
  print(ci)
}
```



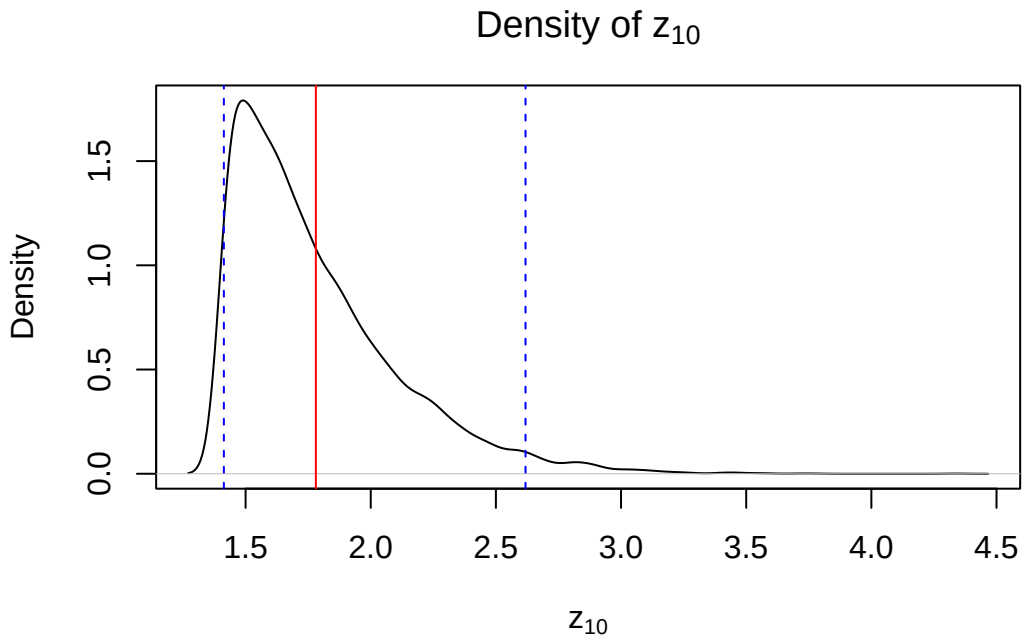

```
    2.5%    97.5%
1.214774 2.502489
```



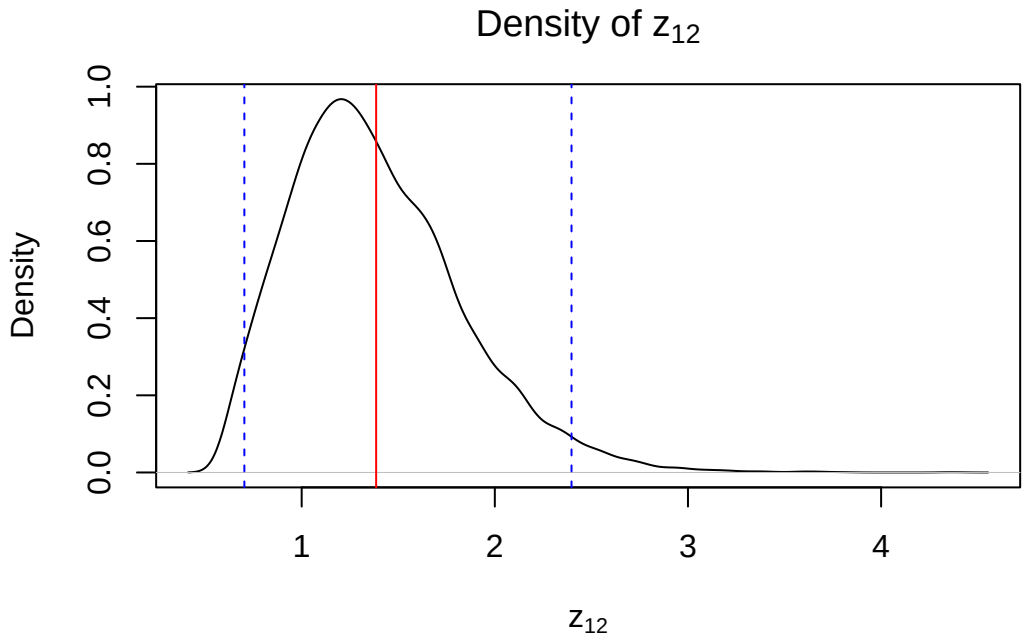

```
    2.5%    97.5%
1.709726 2.741507
```



```
    2.5%    97.5%
2.007599 2.968533
```




```
    2.5%    97.5%
1.413184 2.618399
```




```
     2.5%     97.5%
0.7032876 2.3971134
```

```
par(mfrow = c(1, 1))
```

When we change either $a$ or $b$, the posterior densities change in both cases. When we change $b$ from 1 to 100, the prior favors smaller values of the rate parameter $\theta$ (posterior mean decreases) and the posterior means of the censored values increase relative to Part C. When we increase $a$ from 1 to 100, the opposite occurs: prior favors larger values of the rate parameter $\theta$ (posterior mean increases) and the posterior means of the censored values decrease relative to Part C. The shape of the density for theta remains the same in both cases, but the posterior densities for the censored values become less skewed when we increase $b$ and more skewed when we increase $a$. This occurs because the expected lifetime under the Gamma model is $r/\theta$. The observed values are also constrained to exceed their respective censoring times.